

Borrowed size and borrowed administrative power: Effects of high-speed rail network on industrial upgrading and variegated externalities in the Yangtze River Delta, China

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ABSTRACT

While the externalities of urban size on regional transportation infrastructure effects have been extensively discussed, the role of urban administrative power has received comparatively less scholarly attention. This study delves into the impact of high-speed rail (HSR) development on urban industrial upgrading and regional development, focusing on both economic and administrative externalities. By applying the concepts of ‘borrowed size’ and ‘agglomeration shadow,’ we analyze the effects of HSR on industrial upgrading in the Yangtze River Delta (YRD) region using panel data from 199 county-level units between 2000 and 2017. Employing multi-stage difference-in-differences (DID) and difference-in-difference-in-differences (DDD) models, our findings reveal that HSR significantly catalyzes industrial upgrading in the YRD, with a notable reduction in the output value of the secondary sector by 4.4 % and an expansion of the tertiary sector by 7.4 %. Additionally, this study indicates a positive correlation between urban size and manufacturing sector exit, while administrative power enhances the service sector’s concentration in HSR-connected regions. These findings not only contribute to the existing body of scholarship on HSR’s role in industrial upgrading but also advance a novel methodological approach that bridges transport geography and urban political economy through the lens of transport infrastructure as a mediating mechanism.

1. Introduction

Large-scale transport infrastructure, particularly the high-speed rail (HSR) systems, has long fascinated scholars in transport geography and regional studies due to its role in driving socioeconomic transformation. Recent scholarly discourse has placed increasing emphasis on the role of externalities in shaping transportation development, particularly in fostering urban growth and city-regional formations (Wetwitoo and Kato, 2017). Studies have conceptualized HSR effects as a hybrid of two externalities: agglomeration externalities, viewing HSR stations and routes as critical infrastructure projects, and network externalities, where HSR construction enhances regional transportation networks (De

Rus and Nombela, 2007; Bracaglia et al., 2020; Tang et al., 2021). These findings have justified continued investment in regional transport infrastructure, such as the inter-regional Shinkansen in Japan, the cross-national Eurostar, and the railway lines connecting China with South-east Asia. Nonetheless, a growing body of scholarship has unfolded the complex and diversity about the externalities of HSR effects, broadening the research perspective from mere “agglomeration spillovers” to concepts such as “borrowed size” and “agglomeration shadow” (Burger et al., 2015; Meijers and Burger, 2017). These frameworks illuminate how peripheral smaller cities can capitalize on the labour and consumer markets of neighbouring megacities to facilitate their industrial upgrading, while potentially also serving as catalysts for the industrial

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transformation and development of central cities.

While economic geography often views agglomeration effects through the lens of agglomeration economies, emphasizing the mutual causality between urban economic scale and HSR construction (Zhou and Zhang, 2021; Chang and Zheng, 2022), urban political economy literature suggests that regional infrastructure development and its effects largely depend on local government decision-making and inter-governmental bargaining (Zhang et al., 2020b). Higher-ranked city governments possess greater influence in determining HSR routes and station locations, potentially leveraging HSR construction to shape economic development and regional patterns. Concurrently, research has shown that lower-ranked cities can utilize cross-border transport infrastructure to lease administrative and resource allocation capabilities from nearby higher-ranked cities, thereby integrating their strategies for land development and industrial transformation into regional integration processes (Vickerman, 2015; Qin, 2017; Liang et al., 2020). These changes, in turn, promote cross-border governance and administrative integration (Zhang et al., 2023). This indicates that HSR externalities depend not only on urban scale and agglomeration economies but also on the administrative capacity of city governments. Given the increasing focus on the intersection of economic geography and urban political economy (Wang et al., 2024), we explore the extent to which urban size and administrative power influence the socioeconomic effects of HSR network development in externality formation. We argue that HSR effects exhibit characteristics of both “borrowed size” and “borrowed administrative power,” with heterogeneity across different industrial sectors.

We conduct this study by focusing on China's YRD region, one of the country's most economically developed areas with the highest density of HSR networks, encompassing Shanghai Municipality, Jiangsu Province, Zhejiang Province, and Anhui Province. In China, the introduction of HSR stands out as a potent catalyst for urbanization, economic growth, and industrial evolution (Chen, 2012; Vickerman, 2018). Previous studies have demonstrated the multiscale impacts of HSR construction on urban growth and industrial agglomeration in this region, elucidating intense inter-city competition and cooperation in the YRD region, which profoundly influences HSR construction and its effects (Dai et al., 2018; Diao, 2018; Shao et al., 2017; Wang and Duan, 2018; Wang et al., 2018).

Using county-level panel data from the YRD from 2000 to 2017, we analyze HSR effects on industrial structure and examine whether these effects vary due to externalities of urban size and administrative power. We employ a multi-stage difference-in-differences (DID) model to analyze the causal relationship between HSR and socioeconomic activities, treating HSR projects as continuous policy shocks. To mitigate bias from sample selection and model estimation, we use Propensity Score Matching (PSM) (Liang et al., 2020). We also design a difference-in-difference-in-differences (DDD) model to reveal the influence of different externalities, including agglomeration economies and administrative power, on HSR effects on industrial upgrading. Furthermore, considering the complexity and heterogeneity of HSR externalities, we focus on industrial structure, disaggregating it into secondary and tertiary industries.

The paper is structured as follows: Section 2 offers a literature review, culminating in three research hypotheses. Section 3 delineates our empirical approach, encompassing DID, multi-stage DID, and DDD methodologies. Section 4 unveils results and insights, while Section 5 corroborates findings through robustness tests. We conclude by encapsulating key insights and policy implications for HSR-induced industrial upgrading.

2. Literature review and research hypothesis

2.1. HSR effects on urban industrial upgrading and regional spatial structure

Extensive research has explored the diverse effects of HSR

development, encompassing migration (Wang et al., 2019; Hiramatsu, 2018), fixed asset investment (Qin, 2017; Zhang et al., 2019), manufacturing transfer (Xiao et al., 2020), service industry aggregation (Shao et al., 2017), among others. Scholars have identified several mechanisms through which HSR influences socio-spatial evolution, such as spillover effects, corridor effects, and time-space compression (Chen, 2012; Dai et al., 2018; Gutiérrez et al., 1998; Huang and Du, 2020; Vickerman et al., 1999; Wang et al., 2014; Shaw et al., 2014).

Traditional economic geography and location theories explain these effects by focusing on reduced travel time and costs. They posit that HSR promotes intra-regional factor mobility, fostering spillover effects and regional balanced development (Chen and Hall, 2011; Han et al., 2012). For instance, Shen et al. (2014) found that HSR reduced average travel times in Madrid, which positively impacted urbanization and economic development in cities along HSR corridors. Similarly, through case studies of Córdoba and Zaragoza in Spain and Lille in France, Ureña et al. (2009) found that HSR operations would drive population and industrial outflow from large cities, while small and medium-sized cities would experience population inflow and industrial growth.

On the other hand, some studies also revealed increasing regional inequalities resulting from transportation infrastructure development, challenging the traditional location theory. For example, studies by Song et al. (2015) and Diao (2018) discovered that large Chinese cities experienced better improvements in accessibility and industrial upgrading after HSR construction, while some small and medium-sized cities saw no significant improvements in accessibility and even experienced adverse effects. New Economic Geography offers an alternative perspective, rooted in the concepts of agglomeration externalities and increasing returns to scale (Krugman, 1991; Blanquart and Koning, 2017). This approach suggests that economic activities tend to cluster due to positive externalities such as technology, factor endowments, and information sharing. Regions with concentrated production activities may exhibit higher growth rates due to increasing returns to scale. Furthermore, this perspective interprets regionally balanced development facilitated by transportation infrastructure as a crowding-out effect triggered by excessive agglomeration in central cities, rather than a simple spillover effect (Blanquart and Koning, 2017; Vickerman, 2018).

Despite these explanations providing analytical frameworks for HSR effects, two significant challenges persist. First, an increasing number of studies have found that regional spatial evolution is not merely characterized by agglomeration effects centred on major cities, but also by network effects arising from interactions among peripheral small and medium-sized cities - an aspect often underestimated in previous studies (Zhang et al., 2020b; Zhang et al., 2024). Second, industrial upgrading and HSR construction are considered to have potential endogeneity problems (Gao and Zheng, 2020; Zheng and Kahn, 2013). For instance, it is challenging to determine whether HSR affects local urban growth and industrial upgrading, or whether regions with inherent economic growth potential are more likely to be selected for HSR construction. The following sections will elaborate on these two limitations and subsequently propose three hypotheses for this study.

2.2. Externality of HSR effects and borrowed size

Existing research predominantly focuses on the dynamics of central cities, interpreting HSR-driven regional pattern evolution primarily as a regional-scale manifestation of industrial transfer from these central cities (Cheng et al., 2015; Shao et al., 2017; Liang et al., 2020). This central city-centric approach, however, inadvertently overlooks the agency and interactions of peripheral small and medium-sized cities, potentially leading to an incomplete understanding of regional dynamics. Recent studies have revealed that small peripheral cities may not only actively participate in, but also perform remarkably well within regional evolution processes, sometimes outpacing the benefits accrued by central large cities (Xu and Yeh, 2013; Liang et al., 2020). Debate persists over whether HSR network externalities provide opportunities

for smaller cities to leverage the market scale of large cities, or if they face more intense competition and lose development momentum.

To address this gap, regional studies literature has begun to employ the notion of ‘borrowed size’, introduced by [Alonso \(1973\)](#), which demonstrates that small cities can benefit from the agglomeration economies of nearby large cities. ‘Borrowed size’ represents the inconsistency between urban size and its performance within a regional system, particularly emphasizing how small cities can benefit from the “borrowed” agglomeration economies of nearby large cities ([Meijers and Burger, 2017](#)). Conversely, the concept of “agglomeration shadow” suggests that larger cities may constrain growth in neighbouring smaller cities, resulting in fewer functions and lower performance levels than expected given their size ([Burger et al., 2015](#)).

As urban systems evolve from hierarchical to networked structures, regional studies scholars increasingly apply the concept of borrowed size to explore the role of small and medium-sized cities in regional pattern evolution ([Burger et al., 2015](#); [Meijers and Burger, 2017](#); [Volgmann and Rusche, 2020](#)). In hierarchical urban systems, the agglomeration externalities of specific cities are typically rooted in urban size, with advanced urban functions, industrial structure, and modernity often positively correlated with larger urban scale. However, in networked urban systems, the impact of agglomeration externalities is gradually being superseded by network externalities ([Alonso, 1973](#); [Meijers and Burger, 2017](#)). HSR, as a major regional transportation infrastructure, essentially transforms hierarchical urban systems into urban networks. Consequently, HSR effects should be understood through the lens of network externalities, which can manifest as both positive (borrowed size dominating) and negative (agglomeration shadows dominating) processes ([Meijers and Burger, 2017](#)). Inspired by this, we propose our first hypothesis:

Hypothesis 1. HSR-induced industrial upgrading exhibits network externalities, manifesting as either “borrowed size” or “agglomeration shadow”.

2.3. From borrowed size to borrowed administrative power: integrating administrative power into externality analysis

While economic externalities provide insights into the relationship between regional transportation infrastructure, urban development, and regional evolution, analyses focusing solely on economic and market behaviours face scrutiny, particularly regarding the endogeneity issue between HSR construction and economic growth ([Phelps, 2004](#)). One approach to resolving this issue involves employing various econometric methods, including difference-in-differences and instrumental variables, with an increasing number of studies attempting to precisely calculate the net effect of HSR on urban development.

An alternative solution stems from the political economy of infrastructure development. Recent literature emphasizes the need to integrate governmental roles into previously enterprise-centric economic geography analyses ([Peck, 2023](#); [Wang et al., 2024](#)). This shift is similarly reflected in transportation geography and HSR-related research. Scholars focusing on the political economy of infrastructure have found that HSR construction is not entirely determined by businesses and economic activities, but largely depends on local government intentions and negotiations between different levels of government ([Zhang et al., 2023](#); [Wu, 2016](#); [Zhang et al., 2024](#); [Xu and Yeh, 2013](#)).

Specifically, some scholars argue that higher-ranking city governments possess greater administrative power to influence HSR construction details, such as route planning, station location, and train frequency. It leads to HSR development to better serve higher-ranking cities compared to surrounding cities of lower administrative status ([Wang et al., 2018](#)). Conversely, others argued that due to the presence of higher-level regional governments, HSR construction may benefit small and medium-sized cities more. For instance, [Zhang et al. \(2020b\)](#)’s investigation of HSR in China’s Pearl River Delta revealed that differing

intentions between regional and local governments resulted in the most developed large cities potentially possessing the worst station accessibility, while small and medium-sized cities gained superior accessibility. Although studies have identified the potential influence of administrative power on HSR effects, few have conducted detailed investigations comparing the similarities and differences between economic scale and administrative power in influencing HSR outcomes.

This perspective also offers a novel approach to addressing the endogeneity between HSR effects and economic development, suggesting that HSR effects are not merely economic considerations but also involve significant political factors. Consequently, we posit our second research hypothesis:

Hypothesis 2. HSR effects on industrial upgrading will manifest diverse externalities, which stem from city size and administrative level.

Furthermore, the diverse and sometimes contrasting externalities of HSR effects on different industrial sectors have been widely confirmed ([Blanquart and Koning, 2017](#); [Vickerman, 2018](#)). Following HSR network construction, central cities tend to attract more service industries from surrounding areas, while manufacturing industries simultaneously disperse from the core to the periphery ([Lin, 2017](#); [Tang et al., 2021](#); [Zhou and Zhang, 2021](#)).

Traditional economic geography explains this phenomenon through the agglomeration effects brought by HSR access, which elevate land and housing prices, leading enterprises with varying rent sensitivities to adopt different locational strategies ([Zhang et al., 2021](#)). Specifically, manufacturing industries characterized by low output per unit land area and labour intensity are likely to be displaced from central cities, while capital-intensive and knowledge-intensive tertiary industries may more prominently agglomerate towards central cities ([Yu et al., 2019](#)). New Economic Geography concurs that increased urban scale, and rising rents can displace secondary industries but argues that this crowding-out effect also impacts tertiary industries. The divergent evolution of these two sectors stems from the varying intensities of agglomeration and crowding-out effects produced by equivalent urban scales.

While coupling agglomeration and crowding-out effects provides a more comprehensive economic explanation for regional evolution, political economy literature has discovered that regions with similar economic patterns can exhibit entirely different evolutionary trajectories post-HSR construction. For instance, [Tang et al. \(2021\)](#) compared Beijing-centred and Shanghai-centred city-regions in China, finding that post-HSR construction, Beijing could formulate various regional spatial strategies to attract tertiary industry concentration from surrounding areas, while Shanghai lacked this administrative intervention capacity, preventing it from attracting tertiary industries as prominently as Beijing. Conversely, benefiting from regional integration brought by HSR construction, cities surrounding Shanghai, such as Suzhou, could leverage Shanghai’s public services and infrastructure to attract more infrastructure and manufacturing investments.

While coupling agglomeration and crowding-out effects provides a more comprehensive economic explanation for regional evolution, political economy literature has revealed that regions with similar economic patterns can exhibit entirely different evolutionary trajectories post-HSR construction.

Hypothesis 3. HSR effects exhibit heterogeneous externalities across industrial sectors, with the secondary sector potentially experiencing economic “borrowed size” effects, while the tertiary sector may face “agglomeration shadows” stemming from administrative power.

3. Methodology

3.1. Study area and data source

This research delves into the Yangtze River Delta (YRD) region, which is recognized for its early and robust HSR network in China

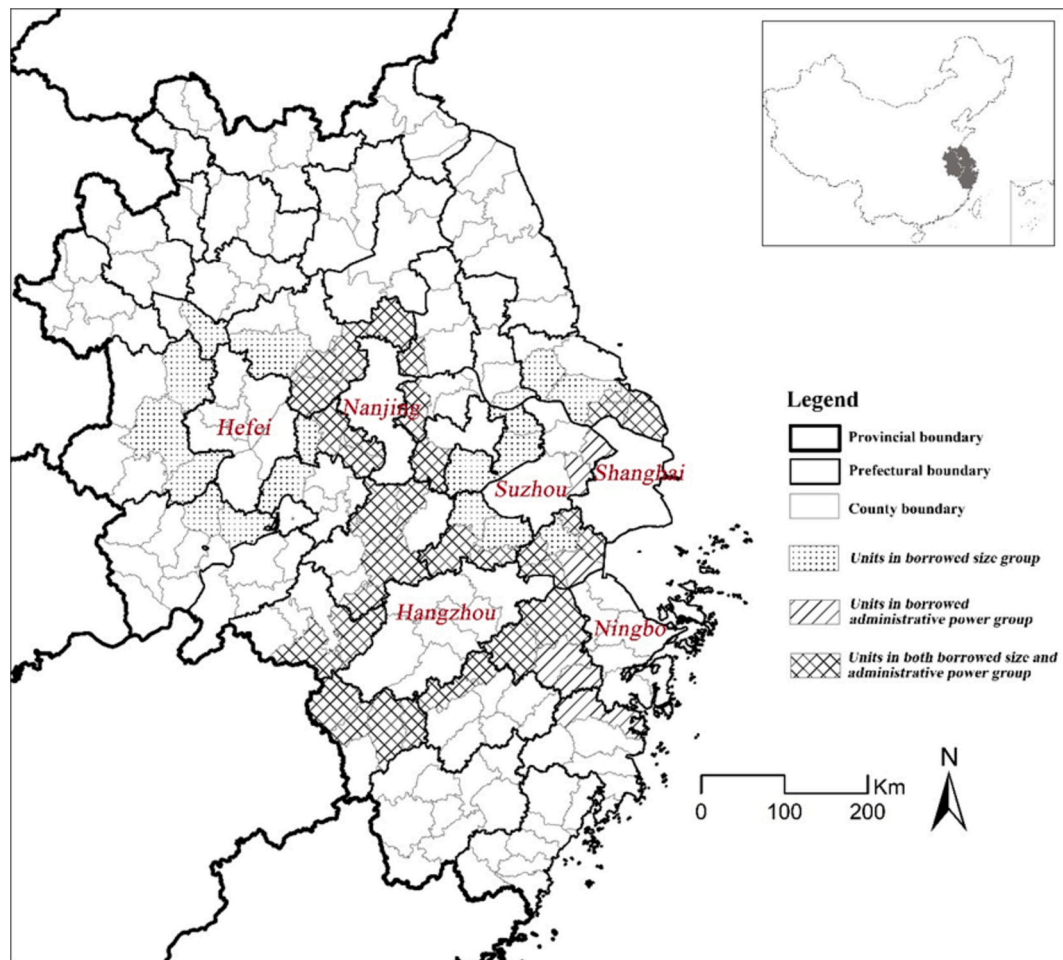


Fig. 1. Study area and units potentially affected by agglomeration and administrative externality.

(Fig. 1). Spanning 358,000 km², the YRD region reported a staggering GDP of 23.72 trillion yuan by 2019 and houses 227 million people, constituting 2.2 %, 23.9 %, and 19.2 % of China in terms of area, GDP, and population, respectively. The YRD, as China's premier economic hub, has spearheaded significant shifts in industrial structure (Tian et al., 2011). Notably, it initiated China's first HSR line, leading the nation in HSR advancements in terms of lines and station density.

To elucidate the diverse externalities stemming from HSR effects as posited in our research hypotheses, we have identified potential zones where these phenomena manifest, focusing on two distinct facets of externalities. The “borrowed size” effect is captured through population metrics, with five cities in the YRD region - Shanghai, Nanjing, Hangzhou, Suzhou, and Hefei - standing out due to their populations exceeding 3 million by 2013. County-level units are proximate to these cities' jurisdictions are earmarked as the “borrowed size group” within our DDD model framework, representing centres of externalities. Comparatively, the “borrowed administrative power” concept is reflected in the administrative echelons of city governance. Cities like Shanghai, Hangzhou, Nanjing, and Ningbo, operating at vice-provincial or higher administrative tiers, wield substantial autonomy in resource allocation, urban planning, and industrial policymaking. County-level units contiguous to these four administrative centres are designated as the “borrowed administrative power group” (Fig. 1), representing zones of administrative power externalities. This dual categorization allows us to examine the differential impacts of economic scale and administrative power on HSR-induced industrial structural changes.

Delving into the intricacies of HSR construction and its consequential impact on industrial structural metamorphosis, this study casts its net

wide, spanning 199 county-level units within the YRD from 2000 through 2017. Such a temporal ambit, encapsulating nine pivotal years pre- and post-2008 HSR inception, is deliberate. Firstly, it furnishes the DID model with a robust foundational period, fortifying against the lurking threat of divergent trends. Secondly, the study period deliberately excludes the years following 2018, when the Chinese central government introduced the YRD integration strategy. This policy represents a significant shock to the industrial structure development in the YRD region (Zhen et al., 2023), which could introduce additional confounding factors to the HSR effects and is therefore not included in the study period.

In terms of analytical scale, we have chosen county-level units as the objects of analysis, rather than prefecture-level units. This choice not only presents a finer granularity but also fully considers the characteristics of HSR deployment and the structure of administrative power. Chinese HSR deployment uses cities and counties as basic units (Qin, 2017). Estimating HSR effects at the prefecture-level may introduce substantial errors. Furthermore, Chinese county governments possess independent fiscal systems and the power to formulate spatial strategies, implying their capacity to negotiate with city and regional governments regarding HSR construction.

Regarding the data source, the HSR information, including the opening year and city, was collected from the Chinese HSR official website (<https://www.12306.cn/index/>) and a statistics website (<http://www.gaotie.cn/>). The socioeconomic data comes from the statistical yearbooks of counties and cities in the CNKI and EPS data platforms. According to Pearson's correlation analysis, the dependent variables were significantly correlated with other indicators, ensuring the

reasonability of the selected variable sets. The errors involved in the adjustment of administrative divisions were also considered. This study transformed the dependent variables and covariates into a logarithmic form to reduce potential heteroscedasticity. We also tested the variance inflation factor ($VIF < 5$) to avoid multicollinearity.

3.2. Methodology

Various methods have been developed to evaluate the impact of HSR on socioeconomic development, including the difference-in-differences (DID) method, the synthetic control method (Li et al., 2019; Li et al., 2020; Kunimi and Seya, 2021) and instrumental variable methods (Faber, 2014; Banerjee et al., 2012). This study employs the DID method due to its widespread application and its foundation in quasi-experimental design principles. The DID approach distinguishes between treatment groups (cities with an HSR station) and control groups (cities without an HSR station), allowing for the assessment of HSR's impact on specific socioeconomic dimensions across time (pre and post-HSR construction).

Our choice of methodology is further motivated by our research hypothesis, which posits that externalities stemming from city size and administrative tiers influence HSR effects. To incorporate these externalities, we extend the DID framework to include a difference-in-difference-in-differences (DDD) model, as demonstrated by Shangguan and Guo (2022) and Jia et al. (2021). Comparative analyses of the DID and DDD results provide a more comprehensive understanding of HSR mechanisms.

Within the framework of this study, two pivotal explanatory variables are delineated: the operational status of HSR within a specified city and the classification of the city within the “borrowed size group.” These variables undergo scrutiny within the DID and DDD models, respectively, to disentangle their nuanced impacts. The crucial dependent variable γ_{it} capture the industrial transformation of city i in year t , encompassing: (1) the output value of secondary industry, (2) the output value of tertiary industry, and (3) the industrial structure ratio. All data are sourced from official statistical yearbooks. The first salient explanatory variable delves into the HSR construction status, encapsulating whether a city has an operational HSR ($Treat_i$) and the timing of the HSR opening (T_t). $Treat_i$ is a dummy variable representing at least one HSR station in city i . If an HSR station exists, $Treat_i = 1$; otherwise, $Treat_i = 0$. T_t is a dummy variable indicating the year of HSR station operation. If the HSR was operated, $T_t = 1$; otherwise, $T_t = 0$. The crux of this variable framework is HSR_{it} , meticulously engineered to quantify the HSR-induced impact on industrial metamorphosis. Mathematically, it is rendered as $Treat_i \times T_t$. The coefficient τ assumes paramount significance, mirroring the variability in HSR_{it} and thereby elucidating the transformative imprint of HSR on the industrial fabric. Furthermore, parameters β_1 and β_2 encapsulate the differential effects of HSR, pre- and post-operational inception. Concurrently, Cov_{it} stands as a repository for a set of control variables, ensuring a holistic analytical perspective. Variables α and ε_{it} indicate the constant and residual items, respectively. The variable μ_i is the unit-fixed effect used to control urban heterogeneity, and δ_t is the time-fixed effect used to manage the time trend.

$$\gamma_{it} = \alpha + \tau HSR_{it} + \beta_1 Treat_i + \beta_2 T_t + \beta_3 Cov_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (1)$$

Given the continuous opening of HSR stations in cities over time, we refer to the research of Shao et al. (2017) and Deng et al. (2020), establishing the multi-stage DID model to systematically examine the dynamic HSR effects. The multi-stage DID model is as follows:

$$\gamma_{it} = \alpha + \tau_1 HSR_{1it} + \tau_2 HSR_{2it} + \dots + \tau_y HSR_{yit} + \beta_1 Treat_i + \beta_2 T_t + \beta_3 Cov_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (2)$$

Where HSR_{1it} is a dummy variable, and its value is equal to one if year t is the first year of an HSR in city i ; otherwise, it is 0. The same assignment standard was used for HSR_{2it} to HSR_{yit} ; τ_y represents the net

HSR effect on the change in industrial structure in city i in year y . The representations of the other variables are the same as those in Equation (1).

The second pivotal explanatory variable delves into the externalities stemming from the HSR effects on industrial upgrading. This nuanced investigation is steered through the DDD model, as articulated in Eq. (3). Central to this exploration is the establishment of the interaction term $HSR_{it} \times BOR_i$, meticulously designed to interlink the HSR effects (HSR_{it}) with both the “borrowed size” (BS) and “borrowed administrative power” (BAP) dimensions. The variable BOR_i assumes pivotal significance, serving as a binary dummy variable that delineates whether city i is potentially ensnared by the ramifications of either “borrowed size” or “borrowed administrative power”. If city i is situated within the realms of potential “borrowed groups”, $BOR_i = 1$, whereas if not, $BOR_i = 0$.

A significant positive value of τ_B indicates that the HSR effect on industrial development will experience borrowed size (or borrowed administrative power) to county-level units adjacent to the municipal areas. Otherwise, a significantly negative value of τ_B means that the HSR effect will experience agglomeration shadow (or administrative shadow).

$$\gamma_{it} = \alpha + \tau_B HSR_{it} \times BOR_i + \tau_{B1} Treat_i \times BOR_i + \tau_{B1} T_t \times BOR_i + \beta_1 Treat_i + \beta_2 T_t + \beta_3 Cov_{it} + \beta_4 BOR_i + \mu_i + \delta_t + \varepsilon_{it} \quad (3)$$

3.3. Descriptive statistics and propensity score matching

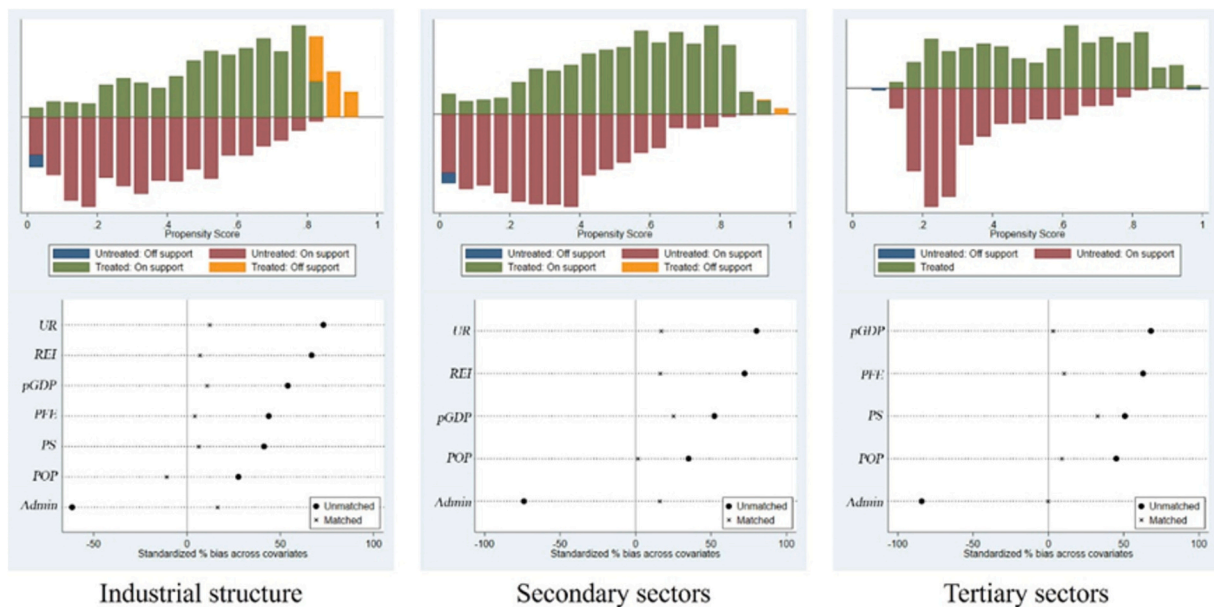
Table 1 outlines the panel data variables central to this study. Our primary dependent variable is industrial structure, represented by γ_{it} , which is the ratio of tertiary to secondary sector output in city i during year t . We also analyze HSR effects on the output magnitudes of both sectors individually to uncover the heterogeneity of HSR effects and their externalities across different industrial sectors, testing Hypothesis 3. To account for potential confounding factors in industrial structure evolution, we incorporate a comprehensive set of control variables based on existing literature: These variables encapsulate various socioeconomic and administrative dimensions that can significantly shape industrial trajectories. Key control variables include urban administrative level (*Admin*), population size (*POP*), and economic development level (*pGDP*), acknowledging the endogeneity between industrial structure evolution and socioeconomic development. We categorize cities into five tiers based on China's administrative hierarchy, ranging from municipality subdistricts to general counties. Following Tang et al. (2021), we examine the influence of population density and economic robustness on industrial shifts. Real estate investments and urbanization rates are considered due to their impact on specific industries such as manufacturing, energy, and construction (Wang et al., 2019). Public financial expenditure (*PFE*) and public service provision (*PS*) are included as they notably promote the tertiary sector. Variables α and ε_{it} indicate the constant and residual items, respectively. The statistical analysis was conducted using Stata 15.

As previously discussed, most socioeconomic activities, including HSR construction, are not entirely random natural experiments but rather quasi-natural experiments with inherent endogeneity in site selection. To address potential endogeneity issues in this quasi-natural experiment, this study employs Propensity Score Matching (PSM), a method that can mitigate bias arising from sample selection and model estimation (Liang et al., 2020). Given our sample size, we implemented one-to-one matching using Nearest Neighbour Matching with replacement. Fig. 2 illustrates the PSM results. The top panels show the distribution of treated and control units for industrial structure, secondary, and tertiary industries. The upper three panels reflect the randomness of the sample and potential errors. Samples labelled “on support” indicate that their inclusion does not significantly affect the randomness and individual characteristics of the treatment and control groups.

Table 1

Definitions and descriptive statistics of variables.

Category	Variables	Abbreviation	Description	Mean	Std. Dev.	Min	Max
Dependent Variables	Industrial structure	<i>IndS</i>	Tertiary/Secondary Output Ratio	0.84	0.35	0.36	4.49
	Secondary Sector Output	<i>Ind2</i>	100 million RMB	218.09	580.86	0.52	9251.4
	Tertiary Sector Output	<i>Ind3</i>	100 million RMB	208.16	884.67	0.85	20,783.47
Treatment Variables	HSR City/County Status	<i>Treat</i>	1 = HSR city, 0 = otherwise	0.41	0.49	0	1
	HSR Opening Time	<i>T</i>	1 = post-HSR, 0 = otherwise	2008.5	5.19	2000	2017
	HSR Effect	<i>HSR</i>	<i>Treat</i> * <i>T</i>	0.13	0.34	0	1
	Cities located in ‘borrowed size’ areas	<i>BS</i>	1 = adjacent to municipal area, 0 = otherwise	0.14	0.34	0	1
Covariates	Cities located in ‘borrowed administrative power’ areas	<i>BAP</i>	1 = adjacent to sub-provincial city, 0 = otherwise	0.07	0.25	0	1
	Administrative level	<i>Admin</i>	City hierarchy (1–5)	4.3	0.91	1	5
	City size	<i>Pop</i>	10,000 residents	101.72	162.68	7.24	2421.47
	Economic development level	<i>pGDP</i>	RMB	33,848.01	33,128.08	1681	218,984.1
	Real estate investment	<i>REI</i>	Billion RMB	57.03	210.44	0	3856.53
	Public financial expenditure	<i>PFE</i>	Billion RMB	59.07	283.64	0.48	7547.62
	Public services provision	<i>PS</i>	Billion RMB	10.02	39.49	0.06	874.1
	Urbanization rate	<i>UR</i>	Proportion of non-agricultural industry	83.7	12.26	37.02	99

**Fig. 2.** The result of the Propensity Score Matching.

Conversely, “off-support” samples suggest unique socioeconomic foundations that are difficult to match in the other group. These samples are subsequently removed from the analysis. Notably, the majority of samples are “on support,” preserving most observations.

To verify the robustness of our matching quality, we conducted balance tests on all matching variables, as shown in the bottom panels. Before matching, there are significant differences between the treatment and control groups. After matching, these differences are substantially reduced, as shown by the closer alignment of points to the diagonal line. This improvement in balance enhances the reliability of our subsequent analysis.

4. Empirical Result

4.1. HSR development and industrial upgrading in the YRD

The development trajectory of the HSR network within the YRD can be delineated into two pivotal phases. Initially, spanning from 2008 to 2013, the HSR infrastructure predominantly linked key regional hubs such as Shanghai, Nanjing, and Hangzhou. This phase laid the foundational architecture for the broader regional HSR network. Subsequently,

from 2013 onwards, the emphasis of the HSR expansion transitioned towards connecting subsidiary cities and inaugurating specialized routes, exemplified by the Huangshan-Hangzhou Tourism Railway. By 2017, a notable 17 HSR lines and 116 stations were operational in the YRD, significantly fortifying economic ties with adjacent regions. As depicted in Table 2, the deployment of new HSR lines progressively gravitated towards serving counties post-2014, signalling a strategic shift from prefectural to county-level connectivity.

Fig. 3 provides an illustrative depiction of the interplay between HSR development and the industrial metamorphosis within the YRD spanning 2000 to 2017. The advent of the HSR catalysed an accelerated spatial reconfiguration of industries. Between 2008 and 2013, urban centres intersecting the HSR nexus witnessed a surge in industrial advancement, exemplified by substantial upticks in tertiary sector contributions, notably in cities like Hangzhou and Hefei. In the subsequent period post-2013, cities integrated within the secondary HSR network (highlighted in yellow) manifested more pronounced shifts in their industrial landscapes, particularly discernible in the southwestern quadrant of the YRD. (See Fig. 4.)

Delving into externalities, discernible variations in the ramifications of HSR development on industrial spillover effects emerge. Notably,

Table 2
Number of newly added cities serviced by HSR in the YRD from 2008 to 2017.

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Number of prefecture-level units	2	5	8	10	3	7	3	11	5	2
Number of county-level units	3	10	6	3	1	5	4	16	5	3

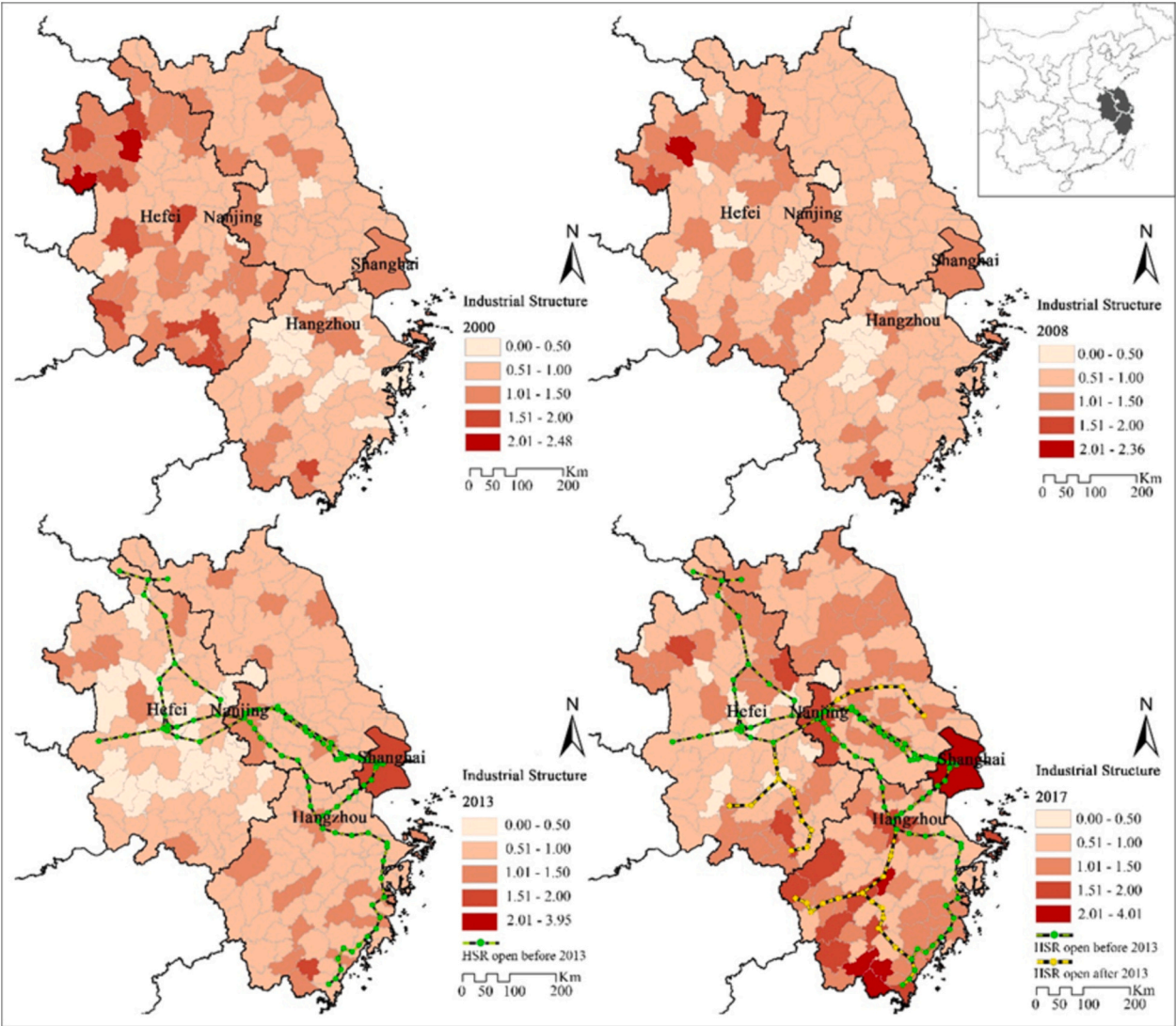


Fig. 3. HSR construction and industrial structure evolution in the YRD in 2000, 2008, 2013 and 2017.

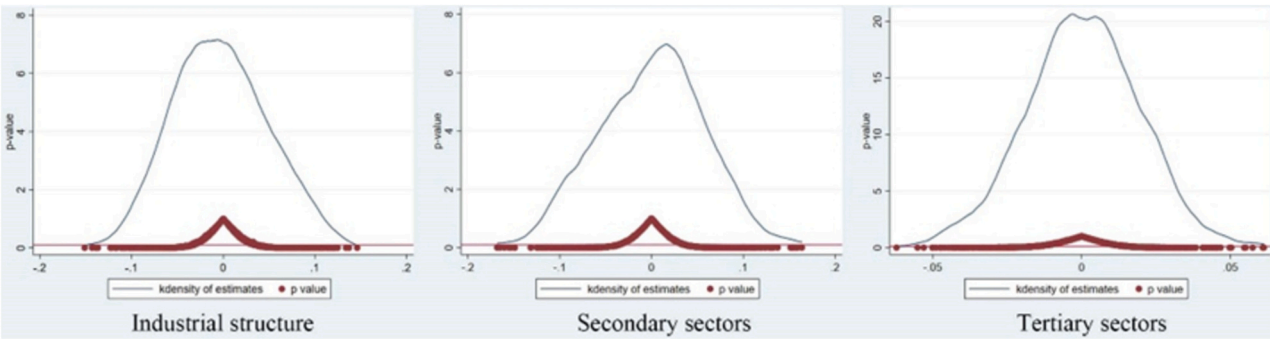


Fig. 4. Results of placebo tests.

locales proximate to Hangzhou and Nanjing consistently showcased enhancements in their industrial fabric, indicative of positive spillover effects catalysing industrial upgrades in these vicinities. Conversely, while Hefei experienced an internal optimization of its industrial composition, peripheral regions exhibited a discernible contraction in the proportion of tertiary sector contributions, signalling potential adverse spillover effects.

4.2. Results from the DID model: HSR effects on industrial upgrading

This segment elucidates the outcomes derived from the DID analysis concerning the influence of HSR on industrial structural evolution, as delineated in Table 3. This study adopted a stepwise regression approach, with results meticulously laid out in Columns 1 through 8. Across all regression iterations, the coefficient associated with HSR consistently registers as positive and statistically significant. The coefficient of HSR in Column 8 is 0.075. This suggests that cities that introduced HSR networks experienced a 7.5 % increase in their industrial structure. Notably, a comparison between Columns 7 and 8 reveals an intriguing pattern: the introduction of public service expenditure as a control variable enhances both the magnitude and statistical significance of the HSR coefficient. This finding suggests a potential interplay between HSR infrastructure and public service provision. Specifically, the industrial upgrading effects of HSR connectivity may not materialize without concurrent investment in public service systems by local governments. Once public service expenditure is controlled for, the impact of HSR construction on industrial upgrading becomes more pronounced. This indicates that the relationship between HSR infrastructure and specific socioeconomic changes is not simply linear but may be contingent upon certain negative confounders or necessary conditions.

Columns 9 and 10 pivot the focus to regression outcomes, with the secondary and tertiary industries serving as the focal dependent variables, respectively. The data intimates a dualistic impact of HSR construction: while it imparts a detrimental effect on the secondary sector, it concurrently fosters growth within the tertiary sector. Based on the coefficients of HSR in columns 9 and 10, which are −0.044 and 0.074, respectively, this dynamic suggests that the drive for industrial structural upgrading, prompted by HSR deployment, functions through a

dual mechanism. Specifically, it diminishes the output of the secondary sector by 4.4 % while stimulating the tertiary sector by 7.4 %. This enhanced connectivity, facilitated by HSR, potentially catalyzes growth within the service, information, and financial sectors. However, it also engenders ancillary repercussions, such as escalating land values and labour costs, thereby exerting a displacement effect on manufacturing and analogous labour-intensive sectors.

Further dissecting the results, the *Treat* variable's consistently positive coefficient accentuates that cities integrated within HSR networks manifest a more pronounced trajectory of industrial advancement relative to their non-HSR counterparts. This observation resonates with the overarching strategic imperatives of the Chinese administration, prioritizing HSR deployment in locales poised for developmental ascendancy.

Among the covariates, the city populace size (*POP*) evinces a salient positive association with the dependent variable, underscoring the potency of agglomeration economies in shaping industrial trajectories. Intriguingly, a converse relationship emerges between cities boasting elevated per capita GDP and economic development metrics and their responsiveness to industrial structural transformation. Such a trend intimates that cities entrenched in advanced developmental trajectories might grapple with path-dependent industrial paradigms, thereby attenuating HSR's transformative impact. Additionally, an unexpected inverse correlation emerges with the administrative echelons of city governance (*Admin*), indicating that cities wielding higher administrative stature exhibit muted responsiveness to HSR-induced industrial dynamics. High-level cities might concurrently undertake other pivotal urban development initiatives and infrastructure projects that dilute the singular impact of the HSR. In summary, HSR-connected cities show more significant industrial upgrading. However, this process involves both growth in the tertiary sector and a simultaneous reduction in the secondary sector, rather than a simple expansion of tertiary industries.

4.3. Results from the DDD model: externalities of HSR effects

Table 4 delves into whether the HSR-induced industrial evolution is shaped by agglomeration (Columns 11–13) and administrative externalities (Columns 14–16). In Columns 11–13, the focus is on the

Table 3
HSR effects on industrial upgrading.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	9(Ind2)	10 (Ind3)
<i>HSR</i>	0.161*** (9.27)	0.151*** (8.74)	0.102*** (5.72)	0.102*** (5.72)	0.056*** (3.19)	0.035* (1.86)	0.029* (1.57)	0.075*** (3.5)	−0.044*** (−2.89)	0.074*** (7.13)
<i>Treat</i>	−0.09* (−1.7)	−0.135** (−2.48)	−0.061* (−1.05)	−0.027 (−0.36)	0.148* (1.67)	0.358*** (3.88)	0.381*** (4.33)	0.566*** (5.97)	0.249*** (2.77)	0.502*** (8.99)
<i>T</i>	−0.035* (−1.89)	−0.028 (−1.55)	−0.014* (−0.77)	−0.014 (−0.77)	0.02 (1.11)	0.025 (1.45)	0.028 (1.61)	−0.016 (−0.8)	−0.011 (−0.69)	−0.031*** (−2.79)
<i>Pop</i>		0.121*** (4.49)	0.036 (1.29)	0.036 (1.29)	0.041 (1.46)	−0.005 (−0.14)	0.053*** (1.49)	0.037* (0.85)	0.595*** (10.84)	0.823*** (26.08)
<i>pGDP</i>			−0.314*** (−11.87)	−0.314*** (−11.87)	−0.241*** (−9.48)	−0.207*** (−7.99)	−0.168*** (−5.65)	−0.319*** (−8.35)	1.037*** (22.68)	0.507*** (18.75)
<i>Admin</i>				−0.033 (−0.56)	−0.172** (−2.39)	−0.314*** (−4.28)	−0.335*** (−4.66)	−0.494*** (−6.55)	−0.15** (−2.02)	−0.467*** (−13.11)
<i>REI</i>					−0.111*** (−12.51)	−0.078*** (−9.1)	−0.07*** (−7.7)	−0.051*** (−5.37)	0.078*** (12.34)	
<i>UR</i>						−0.099*** (−10.69)	−0.091*** (−9.04)	−0.109*** (−9.73)	0.123*** (6.19)	
<i>PFE</i>							−0.127*** (−3.5)	−0.189*** (−3.66)		−0.141*** (−7.39)
<i>PS</i>								−0.021 (−0.7)		0.031*** (3.2)
Constant	1.018*** (26.66)	0.623*** (6.74)	3.62*** (13.2)	3.753*** (8.39)	3.54*** (7.69)	4.11*** (8.57)	3.704*** (7.32)	5.77*** (9.49)	−8.4*** (−11.04)	−3.388*** (−8.24)
County FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
R ²	0.612	0.614	0.639	0.639	0.6677	0.6857	0.6874	0.7214	0.9894	0.994

Symbols *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

Table 4

Externalities of HSR effects on industrial structure.

Variables		Borrowed size			Borrowed administrative power		
		(11) (IndS)	(12) (Ind2)	(13) (Ind3)	(14) (IndS)	(15) (Ind2)	(16) (Ind3)
Before	Control & borrowed	3.71	−10.679	−8.842	3.7	−10.628	−8.931
	Control & N-borrowed	3.818	−10.808	−8.819	3.8	−10.769	−8.907
	Treated & borrowed	3.763	−10.718	−8.829	3.716	−10.669	−8.941
	Treated & N-borrowed	3.737	−10.717	−8.837	3.733	−10.67	−8.913
	Diff(T-C)	0.133*** (5.43)	−0.129*** (6.77)	0.032 (1.59)	0.083*** (3.54)	−0.139*** (7.23)	−0.004 (0.23)
After	Control & borrowed	3.605	−10.811	−8.966	3.687	−10.791	−8.975
	Control & N-borrowed	3.837	−10.996	−8.848	3.805	−10.951	−8.941
	Treated & borrowed	3.867	−10.901	−8.74	3.807	−10.826	−8.901
	Treated & N-borrowed	3.933	−10.875	−8.687	3.932	−10.845	−8.722
	Diff(T-C)	0.166*** (3.61)	−0.212*** (7.18)	0.065 (1.57)	−0.007 (0.14)	−0.141*** (4.26)	−0.145*** (3.76)
HSR × BOR		0.033 (0.62)	−0.082** (−2.34)	0.034 (0.74)	−0.09** (−1.65)	−0.002 (−0.04)	−0.14*** (−3.31)
POP		−0.188*** (−10.451)	0.92*** (90.671)	0.876*** (73.125)	−0.189*** (−10.463)	0.918*** (87.27)	0.88*** (75.229)
pGDP		−0.197*** (−10.576)	1.11*** (87.412)	0.928*** (80.177)	−0.196*** (−10.561)	1.106*** (85.639)	0.934*** (81.949)
Admin		−0.093*** (−9.22)	−0.024*** (−3.911)	−0.111*** (−14.055)	−0.088*** (−8.837)	−0.028*** (−4.531)	−0.101*** (−13.126)
REI		−0.059*** (−3.787)	0.034*** (5.061)		−0.059*** (−3.791)	0.035*** (4.882)	
UR		−0.11*** (−10.432)	0.023*** (3.388)		−0.108*** (−10.195)	0.026*** (3.674)	
PFE		0.238*** (7.627)		0.101*** (8.105)	0.23*** (7.304)		0.094*** (7.282)
PS		0.071*** (2.828)		0.001 (0.067)	0.078*** (3.062)		−0.002 (−0.183)

Symbols *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

agglomeration externality, probing if HSR amplifies industrial structure transitions due to borrowed size. Notably, Column 11 reveals a non-significant coefficient (0.033) for the interaction term $HSR \times BOR$. This suggests that while small and medium-sized cities in the YRD show signs of borrowed size effects in industrial adjustments, it doesn't function in the HSR effect. In more detail, the coefficients of Before-Diff (T-C) and After-Diff (T-C) are 0.133 and 0.166, respectively.

It appears that the way industrial structures evolve in these regions is notably different from other areas, both before and after the introduction of HSR. This suggests that in small and medium-sized cities within the YRD, there's a clear influence of positive agglomeration externality, characterized by 'borrowed size'. However, it indicates that this 'borrowed size' effect doesn't directly correlate with the effect of HSR. Essentially, the industrial upgrading resulting from HSR construction, as indicated in Column 8, is distinct and separate from those driven by agglomeration externality, as seen in Column 11.

Moving to Column 12, a compelling observation emerges: a pronounced crowding-out effect on the secondary sectors due to HSR is amplified in borrowed-size regions. The significant negative coefficient of the interaction term $HSR \times BOR$ could be observed. This heightened displacement effect stems from increased affluence and consumer activity from larger cities post-HSR, escalating costs in adjacent cities and leading to secondary sector attrition. Conversely, Column 13 indicates that the effect on the tertiary sector remains unaffected by agglomeration externality. In sum, this agglomeration externality is primarily manifested in the crowding-out effect of the secondary sectors after the HSR becomes operational. While neighbouring cities see rapid structural shifts, the outcome is a more accelerated decline in secondary sectors rather than a transition to the tertiary, potentially impacting these cities' overall economic growth.

Transitioning to administrative externality, Columns 14–16 scrutinize the influence of administrative power gradients on HSR effects. Column 14 reveals a significant negative coefficient for $HSR \times BOR$, signalling an administrative shadow of the HSR effect on industrial structure. Notably, lower-tier cities around higher administrative hubs

witness subdued industrial upgrading post-HSR compared to their counterparts. Specifically, according to $HSR \times BOR$ coefficients for the secondary sectors in Column 15 and the tertiary sector in Column 16, it is evident that this administrative shadow predominantly stymies tertiary sector growth in these regions, diverting potential progress to higher-tier cities.

In summary, the findings of this study substantiate our three proposed hypotheses. First, following HSR implementation, areas proximate to large cities or high-ranking administrative centres experience significant network externalities. Second, urban characteristics exhibit differential impacts: urban size predominantly affects secondary industries with limited impact on tertiary industries, while urban administrative capacity substantially influences tertiary industries with minimal effect on secondary ones. Third, HSR construction induces heterogeneous industrial effects, notably a crowding-out effect on secondary industries, particularly pronounced in small and medium-sized cities near large urban centres. These smaller cities may "borrow" the population and market scale of larger cities, increasing land values and consequently raising costs for manufacturing establishments, thus intensifying the displacement of secondary industries. Regarding tertiary industries, while HSR generally enhances their attraction to cities, a "siphoning effect" emerges wherein lower-ranked cities lose tertiary sector activities to adjacent higher-ranked cities, creating a "shadow zone" of tertiary industries surrounding higher-ranked cities. In conclusion, post-HSR construction, secondary industries tend to relocate to areas farther from large cities to mitigate HSR-induced land price increases, while tertiary industries concentrate in higher-ranked cities within the HSR network.

5. Robustness analysis

5.1. Parallel trend test

A fundamental prerequisite for employing the DID model is ensuring that the treatment and control groups display consistent trends before

the intervention (Qin, 2017). This is substantiated through a parallel trend assessment. The outcomes of the parallel trend analysis are delineated in Table 5. Specifically, the coefficients denoted as HSR-Bn depict alterations in the dependent variables leading up to the n^{th} year before the initiation of HSR service. If the coefficients of HSR-Bn are statistically significant, it suggests that there might be unforeseen factors influencing the estimation outcomes. Conversely, if these coefficients are not statistically significant, it allows us to discount the influence of factors other than the HSR on industrial upgrading. As Table 5 indicates, all coefficients remain statistically insignificant, underscoring that there were no discernible pre-existing disparities between the treatment and control groups. This confirms the validity of the parallel trend assumption.

Furthermore, our analysis reveals two distinct waves of industrial upgrading following HSR implementation: one occurring in the inaugural year and another emerging in the third year of operation. This dual-wave pattern can be attributed to heterogeneous time-lag effects across different industrial sectors. Specifically, the impact on tertiary industries manifests predominantly in the initial year and the year following HSR operation, while effects on secondary industries become apparent only in the third year. This temporal disparity in response patterns can be explained by the inherent characteristics of these sectors. Tertiary industries, characterized by light assets, demonstrate higher sensitivity to HSR-induced changes such as increased commercial activities. In contrast, secondary industries, being capital-intensive, exhibit a more sluggish response to the cost-benefit alterations brought by HSR connectivity, likely due to considerations of relocation costs and feasibility (Shao et al., 2017).

5.2. Placebo test

Given that the implementation of HSR is an irreversible event, its evaluation hinges on counterfactual assumptions. A placebo test is conducted to ascertain the causal impact of HSR and rule out spurious relationships. Typically, this involves simulating the policy's implementation a set number of years prior and gauging resultant changes in regression outcomes (Shao et al., 2017). Recognizing that HSR deployment transpires progressively rather than instantaneously, traditional placebo testing isn't directly applicable. As an alternative, a fictitious implementation year is randomly chosen for repeated DID analyses to assess consistency in results. If the test still obtains a significant HSR effect on industrial structure change, some unobserved factors affect the

industrial structure, and the regression results above are biased. Otherwise, product robustness could be accepted.

In this study, the placebo test was iterated 1000 times, with the ensuing probability distribution depicted in Fig. 4. Within this figure, the line made by red points signifies the p-values from individual placebo tests, while the blue curve signifies their density. Two pivotal criteria underpin the robustness of the regression outcomes. Firstly, the coefficients predominantly hover around zero, signalling their lack of significance under the placebo assumption. Secondly, a majority of p-values surpass the 0 threshold, underscoring their statistical insignificance across varied fictional implementation timelines. Consequently, Fig. 4 conclusively ascertains the robustness of the empirical findings presented in this study.

6. Discussion and conclusions

6.1. Discussion

The adage “To get rich, build roads first” reflects the efforts of cities, especially peripheral ones, to access transportation infrastructure for economic development and industrial upgrading. As regional transportation infrastructure like HSR, highways, and aviation expands, their impacts should be understood through network structures, not merely hierarchical ones (Wei and Wang, 2020; Wei et al., 2021). This study explores HSR effect on urban industrial upgrading and regional restructuring through an empirical analysis of the Yangtze River Delta, China. Our findings indicate that HSR impact on regional industrial patterns extends beyond central cities' agglomeration or spillover effects. Peripheral cities also dynamically responded to HSR effects, exhibiting characteristics of network effect. Moreover, the network effect stemmed not only from urban economic scale but was also influenced by urban administrative power. The interplay of economic and administrative factors resulted in heterogeneous HSR effects across different sectors.

First, using the multi-stage DID model, we confirm that cities connected to HSR network underwent significant industrial upgrading. HSR accessibility, along with subsequent infrastructure and public service investments, increased tertiary sector activities. This, in turn, elevated real estate investment and land prices, prompting the relocation of the secondary sector to peripheral cities. The finding echoes previous studies on the industrial impact of HSR development in regional China (Dai et al., 2018; Lin, 2017; Shao et al., 2017). Additionally, we utilised the concept of “borrowed size” from network externality theory and DDD modelling to reveal that HSR effects on regional patterns involve not only the dynamics of central cities but also the agency of peripheral cities. Within the HSR network, small and medium-sized cities near large central cities experienced disproportionate crowding-out effects on secondary industries. They may function as residential areas for nearby large cities, leveraging their housing market scale to increase land development and real estate investment, thereby raising land prices and costs to displace manufacturing. A typical example is Jiaxing in the region, an inconspicuous city near Shanghai, which attracted Shanghai commuters through new urban development around its HSR station. Meanwhile, the surge in land prices hindered manufacturers from relocating from Shanghai to Jiaxing (Huang and Du, 2021; Zhen et al., 2023).

This study examined the HSR-induced development through a political economy lens, aiming to understand how both economic scale and administrative power were crucial in shaping HSR effects. The analysis reveals a nuanced picture: areas surrounding cities with higher administrative power experienced more pronounced tertiary sector shadow effects, while the impact on secondary sectors was less evident. This suggests that administrative power primarily influences tertiary industries (such as IT, innovation, and finance) through public service provision, industrial planning, and subsidies. In contrast, secondary industries, being more sensitive to land prices, were primarily driven by

Table 5
Results of the parallel trend test.

Variables	Industrial structure	Secondary sectors	Tertiary sectors
HSR-B4	0.003 (0.13)	-0.025 (-0.76)	0.005 (0.15)
HSR-B3	0.012 (0.6)	-0.042 (-1.05)	0.040 (0.93)
HSR-B2	0.025 (1.31)	-0.023 (-0.61)	0.045 (0.99)
HSR-B1	0.023 (1.07)	-0.047 (-1.4)	-0.014 (-0.29)
HSR-treat	0.045*** (2.62)	0.0769 (1.31)	0.041* (1.94)
HSR-1st	-0.011 (-0.51)	-0.119** (-2.01)	0.026** (2.17)
HSR-2nd	-0.005 (-0.19)	-0.092** (-2.04)	-0.014 (-0.53)
HSR-3rd	0.036** (1.98)	-0.122*** (-2.94)	-0.024 (-1.33)
Constant	6.457*** (3.74)	4.970 (2.13)	-2.510 (-1.26)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Symbols *, **, and *** denote significance levels of 0.1, 0.05, and 0.01, respectively.

economic cost-benefit factors. Nanjing, the capital of Jiangsu Province in the region, exemplifies this dynamic. With its high administrative power, Nanjing leveraged its superior educational and medical facilities after HSR construction to attract high-skilled talent and high-tech industries from surrounding areas. Notably, this occurred without significantly siphoning or crowding out manufacturing in neighbouring small and middle cities (Luo et al., 2004; Yang et al., 2021).

These results indicate that HSR effects on urban development are multifaceted, involving aspects such as transport geography and the political economy of infrastructure development. HSR development enhances inter-city interactions, affecting both economic activities and local government relations. As a result, peripheral regions may need to tailor their development strategies before integrating into HSR networks. Cities that are focusing on tertiary industries, transitioning from manufacturing to services, or promoting real estate development might benefit from the HSR construction. On the other hand, cities aiming to revitalise secondary industries may not gain the maximum benefits from passenger-dedicated HSR.

6.2. Conclusions

This study emphasizes that HSR, as a government-led regional infrastructure initiative, generates not only various regional economic development effects (externalities based on urban size) but also administrative power restructuring (externalities based on urban governmental administrative power). By employing transportation infrastructure as a mediating lens, we offer a novel insight into the increasingly discussed convergence of economic geography and urban political economy (Jonas and Moio, 2018; Wang et al., 2024). By examining the externalities of HSR effects, this paper enhances the previous understanding of the impact of HSR network development and its variations, which has primarily been viewed from the perspective of accessibility and connectivity (Chen and Hall, 2011; Wang et al., 2018; Diao, 2018).

Our findings demonstrate that HSR network development promoted industrial upgrading in cities along its routes. While previous research has established that HSR facilitates the relocation of secondary industries and the concentration of tertiary industries (Blanquart and Koning, 2017; Shao et al., 2017; Tang et al., 2021; Zhou and Zhang, 2021; Chang and Zheng, 2022), our examination reveals nuanced variations based on city size and administrative rank. Central cities in the HSR network may witness the relocation of secondary industries to non-HSR cities, rather than to other HSR cities. Moreover, while HSR access attracted tertiary industry agglomeration to smaller cities, it also exposed them to the siphoning effects of central cities. As a result, although non-central cities in the HSR network may experience industrial upgrading, the total volume of tertiary industries may not necessarily grow. It depends on the ability of peripheral cities to mitigate the siphoning effects of central cities while simultaneously leveraging HSR to attract tertiary industries from surrounding non-HSR areas.

While previous studies often view transport infrastructure development as synchronous with urban economic growth - a perspective particularly prevalent in neoliberal developed economies (Vickerman, 2018) - this approach may have limitations in developing economies, especially those with strong state intervention. With the rise of deglobalization and the consideration of national security, governmental influence on economic and transport development domains may further strengthen. Unlike the market-driven infrastructure development observed in the US, Europe, and Japan (Vickerman, 2015; Diao, 2018), regional transportation network externalities in emerging economies manifest in both economic and administrative dimensions, offering new perspectives on HSR construction and transport policy analysis. Higher-ranked cities, even if not geographically large, possess the authority to intervene in HSR or other regional transportation infrastructure planning to align with their development visions. Conversely, economically large cities may not necessarily obtain

transportation infrastructure commensurate with their sizes. For instance, Suzhou, second only to Shanghai in economic output within the Yangtze River Delta, has been precluded from constructing its airport due to its proximity to Shanghai's Hongqiao and Pudong airports, as well as Nanjing's Pukou airport. Our analysis aligns with the findings of Wang et al. (2020), Zhang et al. (2020a) and Yang et al. (2023), who suggest that the effects of transportation infrastructure in China are hierarchical and subject to scalar politics due to administrative orientation.

This study contributes to the existing literature on political economy of infrastructure development and its induced development. While higher-ranked cities are more likely to secure HSR stations and train services, this does not necessarily guarantee local industrial upgrading and economic growth. Indeed, HSR development can lead to the displacement of secondary industries and may intensify the siphoning effects of higher-ranked cities, potentially hindering regional economic development. These findings have significant implications for regions such as Southeast Asia, Africa, and India, where governments are attempting to restructure economic development, industrial activities, and regional structures through high-speed transport infrastructure construction (Wilmsmeier and Monios, 2016). As our conclusions indicate, HSR effects on regional development depends not only on pre-existing economic geography but also significantly on administrative power structures, offering valuable insights for future transport development in these regions.

While this study offers some insights, it is important to acknowledge several limitations. This analysis primarily focused on the secondary and tertiary industries. As highlighted by Shao et al. (2017), the impacts of HSR can vary significantly within specific subcategories of these industries. Second, as noted by Wang et al. (2018), factors such as the frequency of train services and strategic station placements can introduce notable heterogeneity in HSR effects. Third, this study only focused on the well-developed Yangtze River Delta region in China. The regional specificity also implies that the conclusions have inherent geographical limitations. Given the Yangtze River Delta's advanced urbanization and polycentric structure, the effects of HSR construction in this area may not be directly applicable to regions in China's interior or other countries in the Global South (Jiao et al., 2014). Thus, future research should address these limitations by comparing network externalities across regions with different political and economic conditions. Additionally, incorporating qualitative methods to explore the mechanisms of administrative externalities would provide an alternative view of HSR transportation-induced urban and regional development.

CRediT authorship contribution statement

Dehao Shi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lei Wang:** Writing – review & editing, Validation, Funding acquisition, Formal analysis, Conceptualization. **Xianchun Zhang:** Writing – review & editing, Validation, Conceptualization. **Tao Yu:** Writing – review & editing, Validation, Funding acquisition, Conceptualization.

Declaration of competing interest

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. The author has no conflicts of interest to declare.

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Data availability

Data will be made available on request.

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